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import time, random, numpy as np, torch, torch.nn as nn, torch.nn.funct
from torch.utils.data import DataLoader, Subset
import torchvision, torchvision.transforms as T
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

SEED = 42
def set_seed(s=SEED):
    random.seed(s); np.random.seed(s); torch.manual_seed(s); torch.cuda
set_seed()
device = torch.device("cuda" if torch.cuda.is_available() else "cpu");

QUICK_TEST = False
EPOCHS = 2 if QUICK_TEST else 50
BATCH_SIZE = 128
CIFAR_CLASSES = ['plane','car','bird','cat','deer','dog','frog','horse']

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Device: cuda

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MEAN, STD = (0.4914,0.4822,0.4465), (0.2470,0.2435,0.2616)
train_tf = T.Compose([T.RandomCrop(32,padding=4), T.RandomHorizontalFl
eval_tf = T.Compose([T.ToTensor(), T.Normalize(MEAN,STD)])

full_aug = torchvision.datasets.CIFAR10('./data', train=True, down
full_clean = torchvision.datasets.CIFAR10('./data', train=True, down
test_set = torchvision.datasets.CIFAR10('./data', train=False, down

g = torch.Generator().manual_seed(SEED)
perm = torch.randperm(len(full_aug), generator=g).tolist()
train_set = Subset(full_aug, perm[:45000])
val_set = Subset(full_clean, perm[45000:])

train_loader = DataLoader(train_set, BATCH_SIZE, shuffle=True, num_w
val_loader = DataLoader(val_set, 256, shuffle=False, num_w
test_loader = DataLoader(test_set, 256, shuffle=False, num_w
print(f"train={len(train_set)} val={len(val_set)} test={len(test_set)}.

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train=45000 val=5000 test=10000

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def count_params(m): return sum(p.numel() for p in m.parameters() if p

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@torch.no_grad()
def evaluate(model, loader, criterion=None):
    model.eval(); correct=total=0; loss_sum=0.0
    for x,y in loader:
        x,y=x.to(device),y.to(device); out=model(x)
        if criterion is not None: loss_sum+=criterion(out,y).item()*x.size(0)
        correct+=(out.argmax(1)==y).sum().item(); total+=x.size(0)
    return correct/total, (loss_sum/total if criterion is not None else 0)

def train_model(model, label, epochs=EPOCHS, lr=0.01, weight_decay=5e-5):
    set_seed(); model=model.to(device)
    crit=nn.CrossEntropyLoss()
    opt=torch.optim.SGD(model.parameters(), lr=lr, momentum=0.9, weight_decay=weight_decay)
    sch=torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=epochs)
    h={'train_loss':[], 'val_loss':[], 'train_acc':[], 'val_acc':[], 'epoch_time':[]}
    print(f"\n=== {label} | params={h['params']:,} | epochs={epochs}")
    for ep in range(epochs):
        model.train(); t0=time.time(); rl=rc=rt=0
        for x,y in train_loader:
            x,y=x.to(device),y.to(device); opt.zero_grad()
            out=model(x); loss=crit(out,y); loss.backward(); opt.step()
            rl+=loss.item()*x.size(0); rc+=(out.argmax(1)==y).sum().item()*x.size(0)
        sch.step(); dt=time.time()-t0
        tr_loss,tr_acc=rl/rt, rc/rt
        va,vl=evaluate(model,val_loader,crit)
        h['train_loss'].append(tr_loss); h['val_loss'].append(vl)
        h['train_acc'].append(tr_acc); h['val_acc'].append(va); h['epoch_time'].append(dt)
        print(f" ep {ep+1:02d}/{epochs} train_loss={tr_loss:.3f} acc={tr_acc:.3f} val_loss={vl:.3f} val_acc={va:.3f} epoch_time={dt:.3f}")
    h['avg_epoch_time']=float(np.mean(h['epoch_time'])); return model

def plot_history(hs, title="Training curves"):
    fig,ax=plt.subplots(1,2,figsize=(13,4.5))
    for h in hs:
        ep=range(1,len(h['train_loss'])+1)
        ax[0].plot(ep,h['train_loss'],'--',label=f"{h['label']} train loss")
        ax[0].plot(ep,h['val_loss'],'-',label=f"{h['label']} val loss")
        ax[1].plot(ep,h['val_acc'],'-',label=f"{h['label']} val acc")
    ax[0].set_xlabel('Epoch'); ax[0].set_ylabel('Loss'); ax[0].set_title('Loss')
    ax[1].set_xlabel('Epoch'); ax[1].set_ylabel('Accuracy'); ax[1].set_title('Accuracy')
    plt.tight_layout(); plt.show()

def plot_confusion(model, label=""):
    model.eval(); ys=[]; ps=[]
    with torch.no_grad():
        for x,y in test_loader:
            out=model(x)
            ps+=out.argmax(1).tolist()
            ys+=y.tolist()
    cm=confusion_matrix(ys,ps)
    plt.imshow(cm, cmap=plt.cm.Blues)
    plt.colorbar()
    plt.title(f'Confusion Matrix for {label}')
    plt.xlabel('Actual')
    plt.ylabel('Predicted')
    plt.tight_layout()
    plt.show()

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        ps.append(model(x.to(device)).argmax(1).cpu()); ys.append
cm=confusion_matrix(torch.cat(ys).numpy(), torch.cat(ps).numpy())
fig,axc=plt.subplots(figsize=(7,6))
ConfusionMatrixDisplay(cm,display_labels=CIFAR_CLASSES).plot(ax=axc,
axc.set_title(f'Confusion matrix - {label}'); plt.tight_layout();

def show_first_conv_filters(model, label=""):
    fc=None
    for m in model.modules():
        if isinstance(m,nn.Conv2d): fc=m; break
    if fc is None or fc.in_channels!=3: print("No 3-ch first conv.");
    w=fc.weight.detach().cpu(); w=(w-w.min())/(w.max()-w.min()+1e-8)
    grid=torchvision.utils.make_grid(w[:min(64,w.size(0))],nrow=8,pad=
    plt.figure(figsize=(6,6)); plt.imshow(grid.permute(1,2,0)); plt.a
    plt.title(f'First conv filters - {label}'); plt.show()

def report_test(model,label):
    acc,_=evaluate(model,test_loader); print(f"{label}: TEST accuracy

```

```

class BasicBlock(nn.Module):
    def __init__(self, in_c, out_c, stride=1):
        super().__init__()
        self.conv1=nn.Conv2d(in_c,out_c,3,stride=stride,padding=1,bias=False)
        self.conv2=nn.Conv2d(out_c,out_c,3,stride=1,padding=1,bias=False)
        self.shortcut=nn.Sequential()
        if stride!=1 or in_c!=out_c:
            self.shortcut=nn.Sequential(nn.Conv2d(in_c,out_c,1,stride=1,padding=0,bias=False))
    def forward(self,x):
        out=F.relu(self.bn1(self.conv1(x))); out=self.bn2(self.conv2(out))
        return F.relu(out + self.shortcut(x))

class ResNet(nn.Module):
    def __init__(self, num_blocks, num_classes=10, dropout=0.0):
        super().__init__(); self.in_c=64
        self.stem=nn.Sequential(nn.Conv2d(3,64,3,1,1,bias=False), nn.BatchNorm2d(64))
        self.layer1=self._stage(64, num_blocks[0],1); self.layer2=self._stage(128,num_blocks[1],2)
        self.layer3=self._stage(256,num_blocks[2],2); self.layer4=self._stage(512,num_blocks[3],2)
        self.pool=nn.AdaptiveAvgPool2d(1); self.dropout=nn.Dropout(dropout)
    def _stage(self, out_c, n, stride):
        layers=[]
        for s in [stride]+[1]*(n-1):
            layers.append(BasicBlock(self.in_c,out_c,s)); self.in_c=out_c
        return nn.Sequential(*layers)
    def forward(self,x):
        x=self.stem(x); x=self.layer1(x); x=self.layer2(x); x=self.layer3(x)
        x=self.layer4(x); x=self.pool(x).flatten(1); x=self.dropout(x); return self.fc(x)

def resnet11(dropout=0.0): return ResNet([1,1,1,2], dropout=dropout)
def resnet18(dropout=0.0): return ResNet([2,2,2,2], dropout=dropout)
def resnet34(dropout=0.0): return ResNet([3,4,6,3], dropout=dropout)
print(f"ResNet-11 params: {count_params(resnet11()),}")
print(f"ResNet-18 params: {count_params(resnet18()),}")

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ResNet-11 params: 9,623,882
ResNet-18 params: 11,173,962

```

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r11,h_r11 = train_model(resnet11(), "ResNet-11", lr=0.1)
r18,h_r18 = train_model(resnet18(), "ResNet-18", lr=0.1)
plot_history([h_r11,h_r18], "ResNet-11 vs ResNet-18")
acc_r11=report_test(r11,"ResNet-11"); acc_r18=report_test(r18,"ResNet-18")
plot_confusion(r18, "ResNet-18")
print(f"\n{'Model':12s}{ 'Params':>12s}{ 'TestAcc':>10s}{ 's/epoch':>10s}")
for h,acc in [(h_r11,acc_r11),(h_r18,acc_r18)]:

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print(f"{h['label']:12s}{h['params']:>12,}{acc*100:>9.2f}%{h['avg']
```

```
=== ResNet-11 | params=9,623,882 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=2.156 acc=0.265 | val_loss=1.902 acc=0.302 | 26.9
ep 02/50 train_loss=1.487 acc=0.452 | val_loss=1.330 acc=0.513 | 27.0
ep 03/50 train_loss=1.212 acc=0.560 | val_loss=1.085 acc=0.615 | 28.2
ep 04/50 train_loss=1.017 acc=0.638 | val_loss=0.949 acc=0.659 | 30.4
ep 05/50 train_loss=0.886 acc=0.687 | val_loss=1.050 acc=0.652 | 27.4
ep 06/50 train_loss=0.786 acc=0.724 | val_loss=1.017 acc=0.653 | 28.1
ep 07/50 train_loss=0.694 acc=0.758 | val_loss=0.816 acc=0.722 | 28.6
ep 08/50 train_loss=0.630 acc=0.781 | val_loss=0.624 acc=0.778 | 28.2
ep 09/50 train_loss=0.578 acc=0.798 | val_loss=0.579 acc=0.802 | 27.8
ep 10/50 train_loss=0.544 acc=0.812 | val_loss=0.686 acc=0.762 | 28.2
ep 11/50 train_loss=0.524 acc=0.818 | val_loss=0.665 acc=0.779 | 27.9
ep 12/50 train_loss=0.496 acc=0.830 | val_loss=0.584 acc=0.800 | 27.8
ep 13/50 train_loss=0.474 acc=0.836 | val_loss=0.580 acc=0.803 | 27.8
ep 14/50 train_loss=0.451 acc=0.846 | val_loss=0.518 acc=0.820 | 27.8
ep 15/50 train_loss=0.437 acc=0.850 | val_loss=0.495 acc=0.830 | 27.8
ep 16/50 train_loss=0.422 acc=0.854 | val_loss=0.537 acc=0.816 | 27.8
ep 17/50 train_loss=0.403 acc=0.862 | val_loss=0.702 acc=0.775 | 27.7
ep 18/50 train_loss=0.387 acc=0.867 | val_loss=0.552 acc=0.819 | 27.5
ep 19/50 train_loss=0.369 acc=0.872 | val_loss=0.427 acc=0.852 | 27.8
ep 20/50 train_loss=0.356 acc=0.879 | val_loss=0.599 acc=0.813 | 27.7
ep 21/50 train_loss=0.341 acc=0.883 | val_loss=0.434 acc=0.854 | 27.7
ep 22/50 train_loss=0.330 acc=0.886 | val_loss=0.511 acc=0.838 | 27.8
ep 23/50 train_loss=0.313 acc=0.891 | val_loss=0.443 acc=0.851 | 27.7
ep 24/50 train_loss=0.308 acc=0.894 | val_loss=0.393 acc=0.869 | 27.6
ep 25/50 train_loss=0.287 acc=0.901 | val_loss=0.460 acc=0.847 | 27.7
ep 26/50 train_loss=0.282 acc=0.902 | val_loss=0.418 acc=0.863 | 27.6
ep 27/50 train_loss=0.262 acc=0.909 | val_loss=0.380 acc=0.869 | 27.7
ep 28/50 train_loss=0.249 acc=0.914 | val_loss=0.372 acc=0.875 | 27.6
ep 29/50 train_loss=0.230 acc=0.921 | val_loss=0.363 acc=0.876 | 27.6
ep 30/50 train_loss=0.219 acc=0.924 | val_loss=0.338 acc=0.886 | 27.6
ep 31/50 train_loss=0.204 acc=0.930 | val_loss=0.368 acc=0.875 | 27.7
ep 32/50 train_loss=0.190 acc=0.934 | val_loss=0.323 acc=0.891 | 27.8
ep 33/50 train_loss=0.170 acc=0.941 | val_loss=0.334 acc=0.890 | 27.8
ep 34/50 train_loss=0.157 acc=0.947 | val_loss=0.305 acc=0.898 | 27.7
ep 35/50 train_loss=0.137 acc=0.954 | val_loss=0.311 acc=0.898 | 27.7
ep 36/50 train_loss=0.119 acc=0.960 | val_loss=0.316 acc=0.902 | 27.6
ep 37/50 train_loss=0.106 acc=0.963 | val_loss=0.311 acc=0.906 | 27.7
ep 38/50 train_loss=0.090 acc=0.970 | val_loss=0.272 acc=0.912 | 27.7
ep 39/50 train_loss=0.074 acc=0.975 | val_loss=0.278 acc=0.915 | 27.8
ep 40/50 train_loss=0.058 acc=0.981 | val_loss=0.260 acc=0.922 | 27.6
ep 41/50 train_loss=0.049 acc=0.985 | val_loss=0.274 acc=0.917 | 27.8
ep 42/50 train_loss=0.036 acc=0.989 | val_loss=0.238 acc=0.928 | 27.7
ep 43/50 train_loss=0.029 acc=0.992 | val_loss=0.235 acc=0.930 | 27.7
ep 44/50 train_loss=0.025 acc=0.993 | val_loss=0.239 acc=0.929 | 27.6
ep 45/50 train_loss=0.020 acc=0.995 | val_loss=0.238 acc=0.931 | 27.9
ep 46/50 train_loss=0.018 acc=0.996 | val_loss=0.234 acc=0.933 | 27.8
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ep 47/50 train_loss=0.016 acc=0.997 | val_loss=0.228 acc=0.935 | 27.7
ep 48/50 train_loss=0.013 acc=0.997 | val_loss=0.227 acc=0.934 | 27.6
ep 49/50 train_loss=0.012 acc=0.998 | val_loss=0.229 acc=0.935 | 28.0
ep 50/50 train_loss=0.013 acc=0.997 | val_loss=0.228 acc=0.934 | 27.6

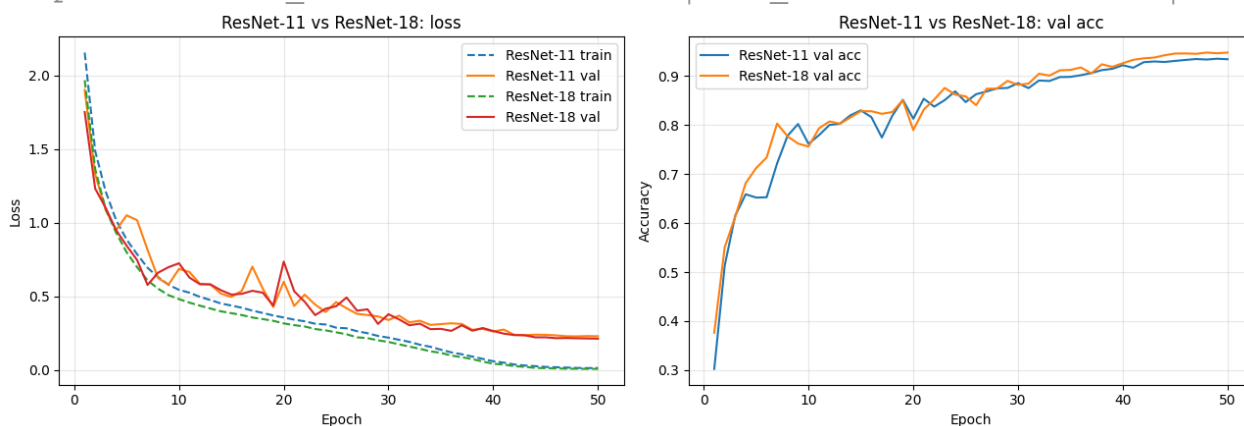
=== ResNet-18 | params=11,173,962 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=1.969 acc=0.289 | val_loss=1.752 acc=0.376 | 40.0
ep 02/50 train_loss=1.365 acc=0.498 | val_loss=1.231 acc=0.551 | 40.3
ep 03/50 train_loss=1.093 acc=0.609 | val_loss=1.102 acc=0.613 | 40.3
ep 04/50 train_loss=0.932 acc=0.669 | val_loss=0.949 acc=0.682 | 40.1
ep 05/50 train_loss=0.799 acc=0.717 | val_loss=0.844 acc=0.712 | 40.1
ep 06/50 train_loss=0.698 acc=0.756 | val_loss=0.746 acc=0.734 | 40.1
ep 07/50 train_loss=0.608 acc=0.789 | val_loss=0.577 acc=0.803 | 40.2
ep 08/50 train_loss=0.552 acc=0.808 | val_loss=0.660 acc=0.777 | 40.1
ep 09/50 train_loss=0.507 acc=0.826 | val_loss=0.698 acc=0.762 | 40.0
ep 10/50 train_loss=0.481 acc=0.834 | val_loss=0.724 acc=0.756 | 40.0
ep 11/50 train_loss=0.457 acc=0.844 | val_loss=0.627 acc=0.793 | 40.0
ep 12/50 train_loss=0.436 acc=0.850 | val_loss=0.582 acc=0.807 | 39.8
ep 13/50 train_loss=0.417 acc=0.857 | val_loss=0.581 acc=0.803 | 39.9
ep 14/50 train_loss=0.398 acc=0.864 | val_loss=0.541 acc=0.816 | 39.9
ep 15/50 train_loss=0.385 acc=0.869 | val_loss=0.511 acc=0.828 | 40.0
ep 16/50 train_loss=0.372 acc=0.874 | val_loss=0.517 acc=0.828 | 40.1
ep 17/50 train_loss=0.356 acc=0.879 | val_loss=0.537 acc=0.823 | 40.0
ep 18/50 train_loss=0.345 acc=0.882 | val_loss=0.524 acc=0.827 | 39.7
ep 19/50 train_loss=0.333 acc=0.886 | val_loss=0.437 acc=0.851 | 39.6
ep 20/50 train_loss=0.316 acc=0.892 | val_loss=0.736 acc=0.790 | 39.7
ep 21/50 train_loss=0.304 acc=0.896 | val_loss=0.534 acc=0.832 | 39.8
ep 22/50 train_loss=0.295 acc=0.900 | val_loss=0.463 acc=0.852 | 40.0
ep 23/50 train_loss=0.277 acc=0.904 | val_loss=0.372 acc=0.876 | 39.9
ep 24/50 train_loss=0.268 acc=0.908 | val_loss=0.416 acc=0.862 | 40.0
ep 25/50 train_loss=0.255 acc=0.912 | val_loss=0.432 acc=0.859 | 40.0
ep 26/50 train_loss=0.242 acc=0.917 | val_loss=0.492 acc=0.841 | 39.9
ep 27/50 train_loss=0.221 acc=0.924 | val_loss=0.402 acc=0.874 | 39.6
ep 28/50 train_loss=0.216 acc=0.926 | val_loss=0.411 acc=0.874 | 39.5
ep 29/50 train_loss=0.200 acc=0.931 | val_loss=0.312 acc=0.890 | 39.8
ep 30/50 train_loss=0.190 acc=0.935 | val_loss=0.378 acc=0.882 | 39.7
ep 31/50 train_loss=0.174 acc=0.941 | val_loss=0.343 acc=0.885 | 39.8
ep 32/50 train_loss=0.158 acc=0.946 | val_loss=0.303 acc=0.905 | 39.9
ep 33/50 train_loss=0.143 acc=0.950 | val_loss=0.314 acc=0.901 | 39.8
ep 34/50 train_loss=0.126 acc=0.957 | val_loss=0.276 acc=0.912 | 39.9
ep 35/50 train_loss=0.115 acc=0.960 | val_loss=0.279 acc=0.913 | 39.8
ep 36/50 train_loss=0.097 acc=0.967 | val_loss=0.265 acc=0.918 | 39.8
ep 37/50 train_loss=0.086 acc=0.970 | val_loss=0.303 acc=0.905 | 39.8
ep 38/50 train_loss=0.073 acc=0.976 | val_loss=0.266 acc=0.924 | 39.9
ep 39/50 train_loss=0.056 acc=0.981 | val_loss=0.284 acc=0.919 | 39.8
ep 40/50 train_loss=0.041 acc=0.987 | val_loss=0.264 acc=0.926 | 39.9
ep 41/50 train_loss=0.036 acc=0.988 | val_loss=0.246 acc=0.933 | 39.8
ep 42/50 train_loss=0.025 acc=0.992 | val_loss=0.236 acc=0.936 | 39.8
ep 43/50 train_loss=0.020 acc=0.994 | val_loss=0.235 acc=0.938 | 39.8
ep 44/50 train_loss=0.013 acc=0.996 | val_loss=0.221 acc=0.943 | 39.9
ep 45/50 train_loss=0.011 acc=0.997 | val_loss=0.221 acc=0.946 | 39.7

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ep 45/50 train_loss=0.001 acc=0.998 | val_loss=0.221 acc=0.946 | 39.5
ep 46/50 train_loss=0.009 acc=0.998 | val_loss=0.215 acc=0.946 | 39.5
ep 47/50 train_loss=0.008 acc=0.998 | val_loss=0.216 acc=0.945 | 39.4
ep 48/50 train_loss=0.008 acc=0.998 | val_loss=0.214 acc=0.948 | 39.7
ep 49/50 train_loss=0.007 acc=0.999 | val_loss=0.213 acc=0.947 | 39.8
ep 50/50 train_loss=0.007 acc=0.999 | val_loss=0.212 acc=0.948 | 39.9

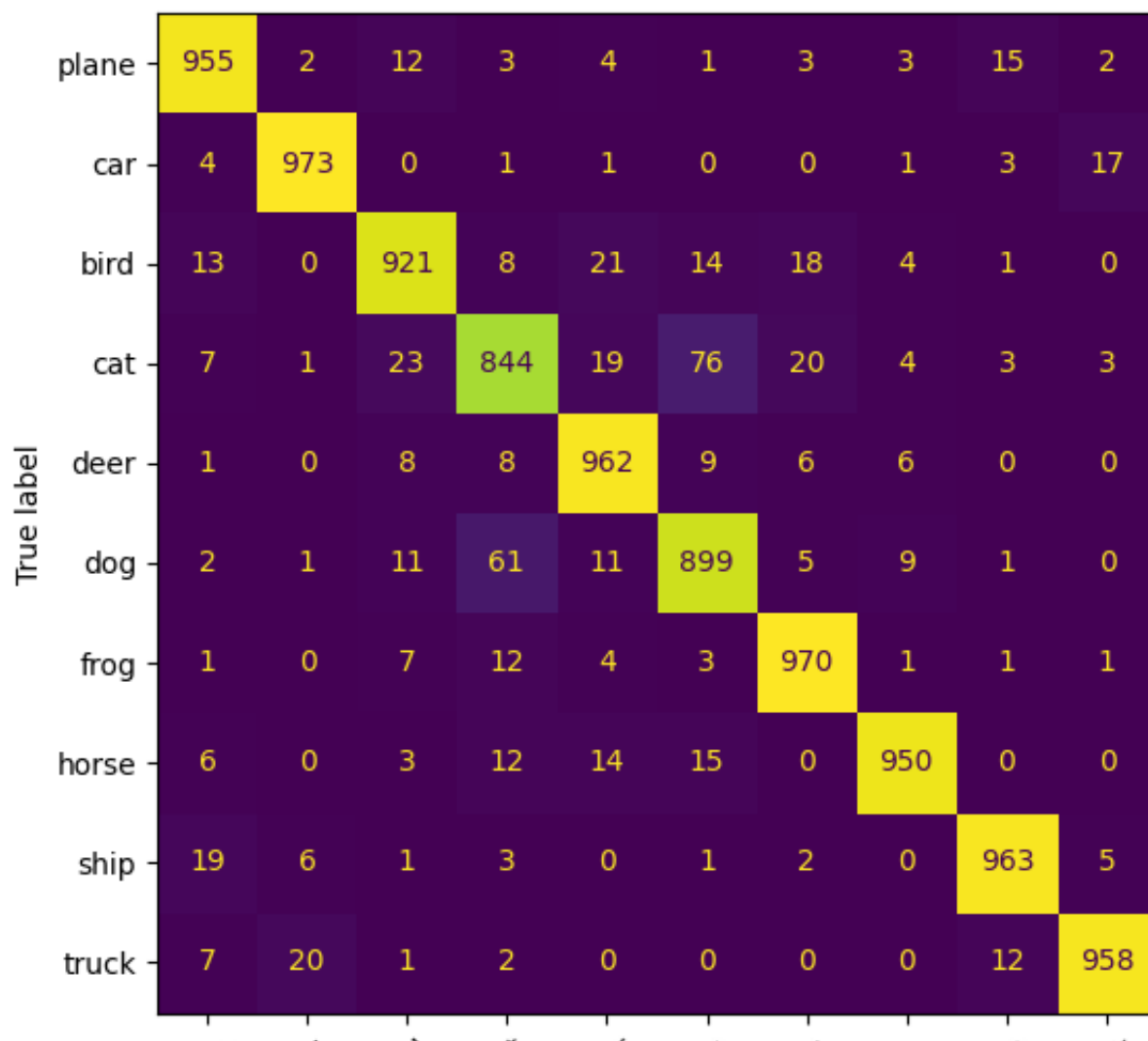
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ResNet-11: TEST accuracy = 92.97%

ResNet-18: TEST accuracy = 93.95%

Confusion matrix — ResNet-18



plane cat bird cat deer dog frog horse ship truck

Predicted label

Model	Params	TestAcc	s/epoch
ResNet-11	9,623,882	92.97%	27.8s
ResNet-18	11,173,962	93.95%	39.9s

```
r11_d3,h11_3=train_model(resnet11(0.3),"ResNet-11 drop=0.3",lr=0.1)
r11_d5,h11_5=train_model(resnet11(0.5),"ResNet-11 drop=0.5",lr=0.1)
r18_d3,h18_3=train_model(resnet18(0.3),"ResNet-18 drop=0.3",lr=0.1)
r18_d5,h18_5=train_model(resnet18(0.5),"ResNet-18 drop=0.5",lr=0.1)
plot_history([h_r11,h11_3,h11_5], "ResNet-11 dropout")
plot_history([h_r18,h18_3,h18_5], "ResNet-18 dropout")
report_test(r11_d3,"ResNet-11 drop=0.3"); report_test(r11_d5,"ResNet-11
report_test(r18_d3,"ResNet-18 drop=0.3"); report_test(r18_d5,"ResNet-18
```

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=== ResNet-11 drop=0.3 | params=9,623,882 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=2.068 acc=0.295 | val_loss=1.613 acc=0.403 | 30.9
ep 02/50 train_loss=1.476 acc=0.459 | val_loss=1.435 acc=0.498 | 33.6
ep 03/50 train_loss=1.234 acc=0.556 | val_loss=1.110 acc=0.600 | 29.3
ep 04/50 train_loss=1.050 acc=0.627 | val_loss=0.980 acc=0.655 | 27.8
ep 05/50 train_loss=0.925 acc=0.672 | val_loss=0.879 acc=0.689 | 27.7
ep 06/50 train_loss=0.845 acc=0.704 | val_loss=1.032 acc=0.635 | 28.2
ep 07/50 train_loss=0.764 acc=0.732 | val_loss=0.772 acc=0.723 | 28.0
ep 08/50 train_loss=0.696 acc=0.759 | val_loss=0.679 acc=0.759 | 27.5
ep 09/50 train_loss=0.626 acc=0.784 | val_loss=0.665 acc=0.768 | 28.5
ep 10/50 train_loss=0.587 acc=0.798 | val_loss=0.636 acc=0.784 | 27.7
ep 11/50 train_loss=0.552 acc=0.811 | val_loss=0.625 acc=0.780 | 28.1
ep 12/50 train_loss=0.525 acc=0.821 | val_loss=0.621 acc=0.780 | 28.0
ep 13/50 train_loss=0.494 acc=0.831 | val_loss=0.578 acc=0.793 | 27.9
ep 14/50 train_loss=0.474 acc=0.837 | val_loss=0.513 acc=0.821 | 27.8
ep 15/50 train_loss=0.453 acc=0.845 | val_loss=0.569 acc=0.803 | 28.2
ep 16/50 train_loss=0.437 acc=0.849 | val_loss=0.497 acc=0.830 | 27.8
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ep 16/50 train_loss=0.427 acc=0.857 | val_loss=0.427 acc=0.857 | 27.9
ep 17/50 train_loss=0.420 acc=0.857 | val_loss=0.569 acc=0.812 | 27.9
ep 18/50 train_loss=0.403 acc=0.863 | val_loss=0.521 acc=0.825 | 27.8
ep 19/50 train_loss=0.388 acc=0.865 | val_loss=0.488 acc=0.833 | 28.3
ep 20/50 train_loss=0.378 acc=0.872 | val_loss=0.495 acc=0.840 | 27.7
ep 21/50 train_loss=0.360 acc=0.876 | val_loss=0.456 acc=0.839 | 27.7
ep 22/50 train_loss=0.346 acc=0.882 | val_loss=0.433 acc=0.854 | 27.7
ep 23/50 train_loss=0.332 acc=0.884 | val_loss=0.449 acc=0.844 | 28.1
ep 24/50 train_loss=0.317 acc=0.891 | val_loss=0.408 acc=0.864 | 27.8
ep 25/50 train_loss=0.305 acc=0.894 | val_loss=0.490 acc=0.835 | 27.7
ep 26/50 train_loss=0.291 acc=0.900 | val_loss=0.433 acc=0.859 | 27.6
ep 27/50 train_loss=0.276 acc=0.905 | val_loss=0.427 acc=0.861 | 27.7
ep 28/50 train_loss=0.260 acc=0.911 | val_loss=0.373 acc=0.872 | 27.8
ep 29/50 train_loss=0.246 acc=0.915 | val_loss=0.370 acc=0.874 | 27.7
ep 30/50 train_loss=0.233 acc=0.919 | val_loss=0.356 acc=0.885 | 27.6
ep 31/50 train_loss=0.212 acc=0.928 | val_loss=0.371 acc=0.879 | 27.9
ep 32/50 train_loss=0.200 acc=0.929 | val_loss=0.363 acc=0.883 | 27.7
ep 33/50 train_loss=0.184 acc=0.937 | val_loss=0.318 acc=0.896 | 27.9
ep 34/50 train_loss=0.163 acc=0.945 | val_loss=0.303 acc=0.900 | 27.7
ep 35/50 train_loss=0.151 acc=0.949 | val_loss=0.299 acc=0.902 | 27.8
ep 36/50 train_loss=0.131 acc=0.955 | val_loss=0.313 acc=0.897 | 27.7
ep 37/50 train_loss=0.116 acc=0.961 | val_loss=0.311 acc=0.903 | 27.8
ep 38/50 train_loss=0.101 acc=0.966 | val_loss=0.290 acc=0.905 | 27.7
ep 39/50 train_loss=0.086 acc=0.972 | val_loss=0.281 acc=0.918 | 27.7
ep 40/50 train_loss=0.070 acc=0.977 | val_loss=0.273 acc=0.919 | 27.9
ep 41/50 train_loss=0.057 acc=0.982 | val_loss=0.265 acc=0.916 | 27.7
ep 42/50 train_loss=0.042 acc=0.987 | val_loss=0.253 acc=0.922 | 27.6
ep 43/50 train_loss=0.035 acc=0.990 | val_loss=0.248 acc=0.926 | 28.0
ep 44/50 train_loss=0.028 acc=0.992 | val_loss=0.240 acc=0.930 | 27.9
ep 45/50 train_loss=0.025 acc=0.993 | val_loss=0.246 acc=0.929 | 27.9
ep 46/50 train_loss=0.021 acc=0.994 | val_loss=0.241 acc=0.931 | 27.8
ep 47/50 train_loss=0.018 acc=0.995 | val_loss=0.239 acc=0.932 | 27.9
ep 48/50 train_loss=0.017 acc=0.996 | val_loss=0.238 acc=0.932 | 27.7
ep 49/50 train_loss=0.015 acc=0.997 | val_loss=0.237 acc=0.933 | 27.9
ep 50/50 train_loss=0.016 acc=0.997 | val_loss=0.236 acc=0.932 | 27.9

=== ResNet-11 drop=0.5 | params=9,623,882 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=2.092 acc=0.302 | val_loss=1.595 acc=0.417 | 28.4
ep 02/50 train_loss=1.436 acc=0.475 | val_loss=1.353 acc=0.529 | 28.2
ep 03/50 train_loss=1.160 acc=0.586 | val_loss=1.028 acc=0.633 | 27.9
ep 04/50 train_loss=0.984 acc=0.650 | val_loss=0.929 acc=0.670 | 28.1
ep 05/50 train_loss=0.853 acc=0.700 | val_loss=0.865 acc=0.703 | 28.1
ep 06/50 train_loss=0.758 acc=0.739 | val_loss=0.929 acc=0.674 | 28.3
ep 07/50 train_loss=0.680 acc=0.767 | val_loss=0.769 acc=0.725 | 28.1
ep 08/50 train_loss=0.620 acc=0.788 | val_loss=0.598 acc=0.790 | 28.5
ep 09/50 train_loss=0.580 acc=0.800 | val_loss=0.632 acc=0.786 | 27.9
ep 10/50 train_loss=0.544 acc=0.813 | val_loss=0.618 acc=0.795 | 28.0
ep 11/50 train_loss=0.519 acc=0.824 | val_loss=0.525 acc=0.819 | 28.1
ep 12/50 train_loss=0.500 acc=0.828 | val_loss=0.583 acc=0.801 | 28.5
ep 13/50 train_loss=0.477 acc=0.838 | val_loss=0.583 acc=0.800 | 28.0
ep 14/50 train_loss=0.458 acc=0.846 | val_loss=0.562 acc=0.810 | 28.0

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```

ep 15/50 train_loss=0.440 acc=0.851 | val_loss=0.505 acc=0.830 | 27.9
ep 16/50 train_loss=0.430 acc=0.854 | val_loss=0.518 acc=0.826 | 28.2
ep 17/50 train_loss=0.412 acc=0.861 | val_loss=0.480 acc=0.835 | 28.1
ep 18/50 train_loss=0.397 acc=0.865 | val_loss=0.490 acc=0.829 | 28.0
ep 19/50 train_loss=0.382 acc=0.871 | val_loss=0.528 acc=0.820 | 28.4
ep 20/50 train_loss=0.372 acc=0.874 | val_loss=0.523 acc=0.825 | 28.1
ep 21/50 train_loss=0.355 acc=0.879 | val_loss=0.474 acc=0.836 | 28.1
ep 22/50 train_loss=0.342 acc=0.883 | val_loss=0.490 acc=0.834 | 28.2
ep 23/50 train_loss=0.328 acc=0.888 | val_loss=0.458 acc=0.846 | 28.1
ep 24/50 train_loss=0.317 acc=0.891 | val_loss=0.500 acc=0.838 | 28.2
ep 25/50 train_loss=0.311 acc=0.893 | val_loss=0.433 acc=0.857 | 27.9
ep 26/50 train_loss=0.292 acc=0.900 | val_loss=0.395 acc=0.863 | 27.9
ep 27/50 train_loss=0.272 acc=0.907 | val_loss=0.542 acc=0.835 | 28.1
ep 28/50 train_loss=0.261 acc=0.910 | val_loss=0.418 acc=0.861 | 28.1
ep 29/50 train_loss=0.246 acc=0.917 | val_loss=0.324 acc=0.893 | 28.0
ep 30/50 train_loss=0.232 acc=0.921 | val_loss=0.354 acc=0.886 | 28.5
ep 31/50 train_loss=0.214 acc=0.928 | val_loss=0.355 acc=0.883 | 28.0
ep 32/50 train_loss=0.202 acc=0.931 | val_loss=0.363 acc=0.879 | 27.6
ep 33/50 train_loss=0.182 acc=0.939 | val_loss=0.354 acc=0.886 | 27.6
ep 34/50 train_loss=0.165 acc=0.943 | val_loss=0.305 acc=0.901 | 28.1
ep 35/50 train_loss=0.157 acc=0.947 | val_loss=0.299 acc=0.906 | 28.2
ep 36/50 train_loss=0.130 acc=0.955 | val_loss=0.340 acc=0.896 | 27.8
ep 37/50 train_loss=0.117 acc=0.961 | val_loss=0.326 acc=0.903 | 27.6
ep 38/50 train_loss=0.103 acc=0.965 | val_loss=0.283 acc=0.913 | 27.5
ep 39/50 train_loss=0.089 acc=0.971 | val_loss=0.277 acc=0.917 | 28.0
ep 40/50 train_loss=0.070 acc=0.977 | val_loss=0.270 acc=0.916 | 28.1
ep 41/50 train_loss=0.058 acc=0.981 | val_loss=0.263 acc=0.923 | 28.5
ep 42/50 train_loss=0.046 acc=0.986 | val_loss=0.235 acc=0.934 | 27.6
ep 43/50 train_loss=0.036 acc=0.989 | val_loss=0.233 acc=0.934 | 27.5
ep 44/50 train_loss=0.028 acc=0.992 | val_loss=0.243 acc=0.933 | 27.6
ep 45/50 train_loss=0.023 acc=0.994 | val_loss=0.242 acc=0.931 | 28.2
ep 46/50 train_loss=0.021 acc=0.995 | val_loss=0.234 acc=0.934 | 28.0
ep 47/50 train_loss=0.018 acc=0.996 | val_loss=0.230 acc=0.935 | 28.0
ep 48/50 train_loss=0.017 acc=0.996 | val_loss=0.231 acc=0.935 | 27.6
ep 49/50 train_loss=0.015 acc=0.996 | val_loss=0.232 acc=0.937 | 28.0
ep 50/50 train_loss=0.015 acc=0.997 | val_loss=0.231 acc=0.937 | 28.0

=== ResNet-18 drop=0.3 | params=11,173,962 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=1.962 acc=0.300 | val_loss=1.571 acc=0.418 | 40.1
ep 02/50 train_loss=1.447 acc=0.472 | val_loss=1.357 acc=0.504 | 40.0
ep 03/50 train_loss=1.206 acc=0.567 | val_loss=1.175 acc=0.576 | 39.8
ep 04/50 train_loss=1.010 acc=0.641 | val_loss=0.988 acc=0.656 | 40.1
ep 05/50 train_loss=0.853 acc=0.700 | val_loss=0.889 acc=0.693 | 40.2
ep 06/50 train_loss=0.745 acc=0.741 | val_loss=0.710 acc=0.749 | 39.9
ep 07/50 train_loss=0.654 acc=0.774 | val_loss=0.710 acc=0.760 | 39.7
ep 08/50 train_loss=0.593 acc=0.797 | val_loss=0.658 acc=0.776 | 40.0
ep 09/50 train_loss=0.544 acc=0.813 | val_loss=0.607 acc=0.794 | 40.2
ep 10/50 train_loss=0.524 acc=0.821 | val_loss=0.632 acc=0.782 | 39.9
ep 11/50 train_loss=0.495 acc=0.833 | val_loss=0.580 acc=0.802 | 39.6
ep 12/50 train_loss=0.474 acc=0.840 | val_loss=0.628 acc=0.786 | 39.6

```

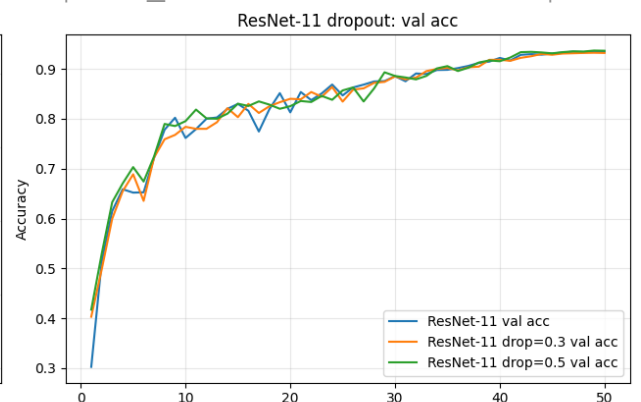
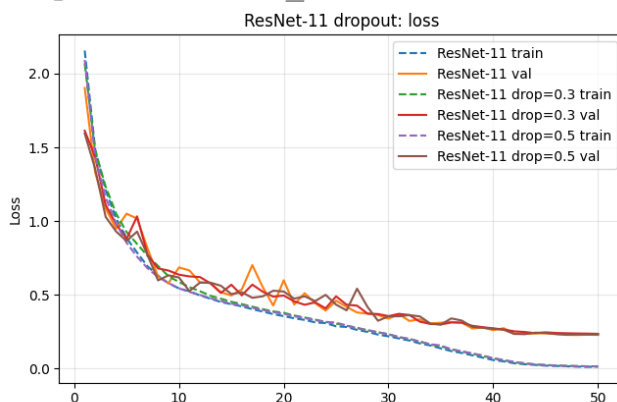
```

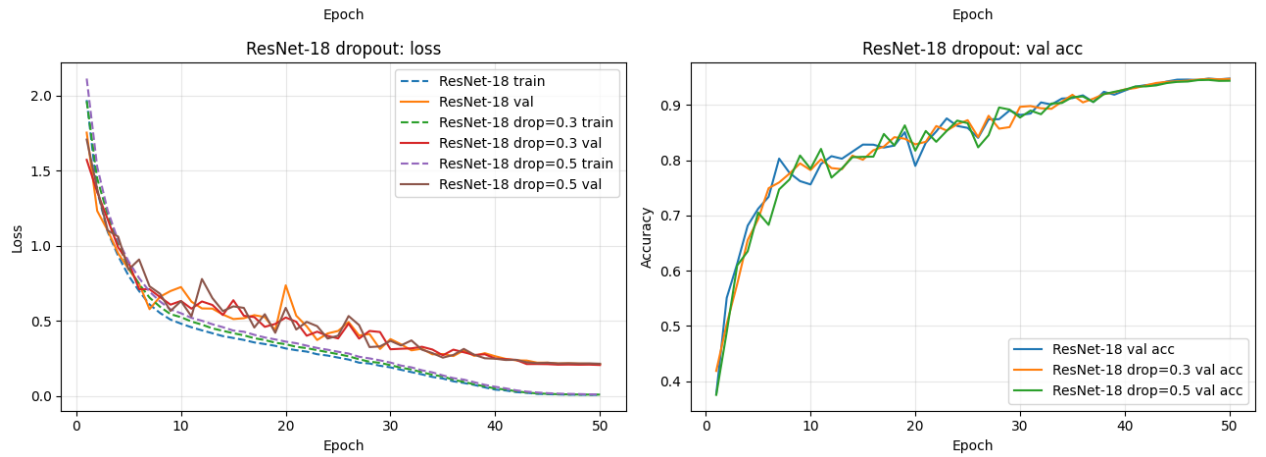
ep 13/50 train_loss=0.449 acc=0.848 | val_loss=0.605 acc=0.784 | 39.8
ep 14/50 train_loss=0.433 acc=0.853 | val_loss=0.538 acc=0.808 | 39.9
ep 15/50 train_loss=0.417 acc=0.858 | val_loss=0.637 acc=0.801 | 40.0
ep 16/50 train_loss=0.403 acc=0.861 | val_loss=0.533 acc=0.818 | 39.9
ep 17/50 train_loss=0.384 acc=0.870 | val_loss=0.524 acc=0.825 | 39.9
ep 18/50 train_loss=0.372 acc=0.872 | val_loss=0.459 acc=0.842 | 39.9
ep 19/50 train_loss=0.354 acc=0.878 | val_loss=0.479 acc=0.839 | 40.0
ep 20/50 train_loss=0.344 acc=0.885 | val_loss=0.522 acc=0.829 | 40.0
ep 21/50 train_loss=0.327 acc=0.890 | val_loss=0.493 acc=0.833 | 39.8
ep 22/50 train_loss=0.318 acc=0.893 | val_loss=0.400 acc=0.862 | 39.7
ep 23/50 train_loss=0.304 acc=0.896 | val_loss=0.427 acc=0.854 | 39.7
ep 24/50 train_loss=0.291 acc=0.902 | val_loss=0.399 acc=0.865 | 40.1
ep 25/50 train_loss=0.278 acc=0.906 | val_loss=0.383 acc=0.873 | 39.5
ep 26/50 train_loss=0.263 acc=0.910 | val_loss=0.481 acc=0.842 | 39.7
ep 27/50 train_loss=0.244 acc=0.917 | val_loss=0.381 acc=0.881 | 39.8
ep 28/50 train_loss=0.227 acc=0.923 | val_loss=0.432 acc=0.857 | 40.0
ep 29/50 train_loss=0.220 acc=0.925 | val_loss=0.425 acc=0.860 | 39.9
ep 30/50 train_loss=0.204 acc=0.930 | val_loss=0.310 acc=0.897 | 39.8
ep 31/50 train_loss=0.188 acc=0.936 | val_loss=0.314 acc=0.898 | 39.8
ep 32/50 train_loss=0.175 acc=0.941 | val_loss=0.316 acc=0.894 | 40.1
ep 33/50 train_loss=0.158 acc=0.945 | val_loss=0.327 acc=0.893 | 39.9
ep 34/50 train_loss=0.140 acc=0.952 | val_loss=0.308 acc=0.906 | 39.9
ep 35/50 train_loss=0.129 acc=0.957 | val_loss=0.270 acc=0.919 | 39.8
ep 36/50 train_loss=0.107 acc=0.963 | val_loss=0.308 acc=0.905 | 39.8
ep 37/50 train_loss=0.093 acc=0.969 | val_loss=0.290 acc=0.912 | 39.9
ep 38/50 train_loss=0.077 acc=0.975 | val_loss=0.272 acc=0.920 | 39.7
ep 39/50 train_loss=0.063 acc=0.979 | val_loss=0.277 acc=0.923 | 39.7
ep 40/50 train_loss=0.049 acc=0.984 | val_loss=0.249 acc=0.928 | 39.7
ep 41/50 train_loss=0.042 acc=0.987 | val_loss=0.244 acc=0.931 | 39.8
ep 42/50 train_loss=0.030 acc=0.991 | val_loss=0.239 acc=0.936 | 40.0
ep 43/50 train_loss=0.022 acc=0.994 | val_loss=0.212 acc=0.940 | 39.9
ep 44/50 train_loss=0.016 acc=0.995 | val_loss=0.212 acc=0.942 | 39.9
ep 45/50 train_loss=0.014 acc=0.997 | val_loss=0.211 acc=0.943 | 39.8
ep 46/50 train_loss=0.011 acc=0.998 | val_loss=0.208 acc=0.944 | 39.7
ep 47/50 train_loss=0.010 acc=0.998 | val_loss=0.208 acc=0.946 | 39.9
ep 48/50 train_loss=0.009 acc=0.998 | val_loss=0.207 acc=0.947 | 39.9
ep 49/50 train_loss=0.008 acc=0.998 | val_loss=0.208 acc=0.947 | 40.0
ep 50/50 train_loss=0.008 acc=0.998 | val_loss=0.205 acc=0.947 | 39.9

=== ResNet-18 drop=0.5 | params=11,173,962 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=2.112 acc=0.260 | val_loss=1.706 acc=0.375 | 39.8
ep 02/50 train_loss=1.525 acc=0.436 | val_loss=1.378 acc=0.489 | 39.7
ep 03/50 train_loss=1.234 acc=0.556 | val_loss=1.102 acc=0.609 | 39.7
ep 04/50 train_loss=1.034 acc=0.632 | val_loss=1.060 acc=0.635 | 39.6
ep 05/50 train_loss=0.895 acc=0.687 | val_loss=0.841 acc=0.705 | 39.7
ep 06/50 train_loss=0.784 acc=0.730 | val_loss=0.908 acc=0.683 | 40.0
ep 07/50 train_loss=0.694 acc=0.762 | val_loss=0.729 acc=0.747 | 39.9
ep 08/50 train_loss=0.628 acc=0.786 | val_loss=0.680 acc=0.765 | 39.8
ep 09/50 train_loss=0.576 acc=0.803 | val_loss=0.565 acc=0.809 | 39.8
ep 10/50 train_loss=0.550 acc=0.814 | val_loss=0.630 acc=0.785 | 39.8
ep 11/50 train_loss=0.519 acc=0.824 | val_loss=0.529 acc=0.821 | 40.0

```

ep 12/50	train_loss=0.500	acc=0.831	val_loss=0.778	acc=0.769	39.9
ep 13/50	train_loss=0.477	acc=0.839	val_loss=0.651	acc=0.785	39.9
ep 14/50	train_loss=0.457	acc=0.845	val_loss=0.564	acc=0.806	40.0
ep 15/50	train_loss=0.434	acc=0.853	val_loss=0.596	acc=0.806	39.7
ep 16/50	train_loss=0.427	acc=0.855	val_loss=0.586	acc=0.807	39.6
ep 17/50	train_loss=0.406	acc=0.863	val_loss=0.455	acc=0.848	39.9
ep 18/50	train_loss=0.389	acc=0.868	val_loss=0.543	acc=0.827	39.9
ep 19/50	train_loss=0.376	acc=0.873	val_loss=0.420	acc=0.863	39.4
ep 20/50	train_loss=0.361	acc=0.879	val_loss=0.586	acc=0.817	39.8
ep 21/50	train_loss=0.350	acc=0.882	val_loss=0.442	acc=0.853	39.9
ep 22/50	train_loss=0.334	acc=0.888	val_loss=0.493	acc=0.833	39.4
ep 23/50	train_loss=0.319	acc=0.892	val_loss=0.464	acc=0.853	39.6
ep 24/50	train_loss=0.307	acc=0.897	val_loss=0.381	acc=0.872	39.9
ep 25/50	train_loss=0.294	acc=0.900	val_loss=0.400	acc=0.867	39.7
ep 26/50	train_loss=0.282	acc=0.906	val_loss=0.531	acc=0.823	39.6
ep 27/50	train_loss=0.260	acc=0.912	val_loss=0.471	acc=0.845	39.9
ep 28/50	train_loss=0.249	acc=0.917	val_loss=0.325	acc=0.896	39.7
ep 29/50	train_loss=0.236	acc=0.919	val_loss=0.329	acc=0.892	39.6
ep 30/50	train_loss=0.222	acc=0.925	val_loss=0.366	acc=0.877	39.7
ep 31/50	train_loss=0.202	acc=0.931	val_loss=0.336	acc=0.890	39.9
ep 32/50	train_loss=0.190	acc=0.936	val_loss=0.369	acc=0.883	39.9
ep 33/50	train_loss=0.170	acc=0.944	val_loss=0.307	acc=0.903	39.9
ep 34/50	train_loss=0.156	acc=0.948	val_loss=0.285	acc=0.904	39.5
ep 35/50	train_loss=0.136	acc=0.954	val_loss=0.254	acc=0.914	39.4
ep 36/50	train_loss=0.118	acc=0.960	val_loss=0.272	acc=0.916	39.7
ep 37/50	train_loss=0.108	acc=0.964	val_loss=0.312	acc=0.906	39.8
ep 38/50	train_loss=0.091	acc=0.970	val_loss=0.273	acc=0.920	39.8
ep 39/50	train_loss=0.073	acc=0.975	val_loss=0.250	acc=0.924	39.7
ep 40/50	train_loss=0.060	acc=0.980	val_loss=0.247	acc=0.928	39.6
ep 41/50	train_loss=0.050	acc=0.984	val_loss=0.239	acc=0.934	39.6
ep 42/50	train_loss=0.037	acc=0.989	val_loss=0.240	acc=0.934	39.9
ep 43/50	train_loss=0.028	acc=0.992	val_loss=0.224	acc=0.936	39.8
ep 44/50	train_loss=0.022	acc=0.994	val_loss=0.218	acc=0.940	39.9
ep 45/50	train_loss=0.018	acc=0.995	val_loss=0.220	acc=0.942	39.7
ep 46/50	train_loss=0.015	acc=0.996	val_loss=0.216	acc=0.943	39.6
ep 47/50	train_loss=0.013	acc=0.997	val_loss=0.217	acc=0.945	39.6
ep 48/50	train_loss=0.012	acc=0.997	val_loss=0.215	acc=0.946	39.6
ep 49/50	train_loss=0.010	acc=0.998	val_loss=0.216	acc=0.944	39.7
ep 50/50	train_loss=0.010	acc=0.998	val_loss=0.214	acc=0.944	39.8





```

ResNet-11 drop=0.3: TEST accuracy = 93.04%
ResNet-11 drop=0.5: TEST accuracy = 93.24%
ResNet-18 drop=0.3: TEST accuracy = 93.52%
ResNet-18 drop=0.5: TEST accuracy = 93.78%
0.9378

```

```

best_alex_acc, best_alex_params = 0.0, 6_975_178
best_vgg_acc, best_vgg_params = 0.0, 9_231_114
finals = [("Best AlexNet", best_alex_params, best_alex_acc),
          ("Best VGG", best_vgg_params, best_vgg_acc)].

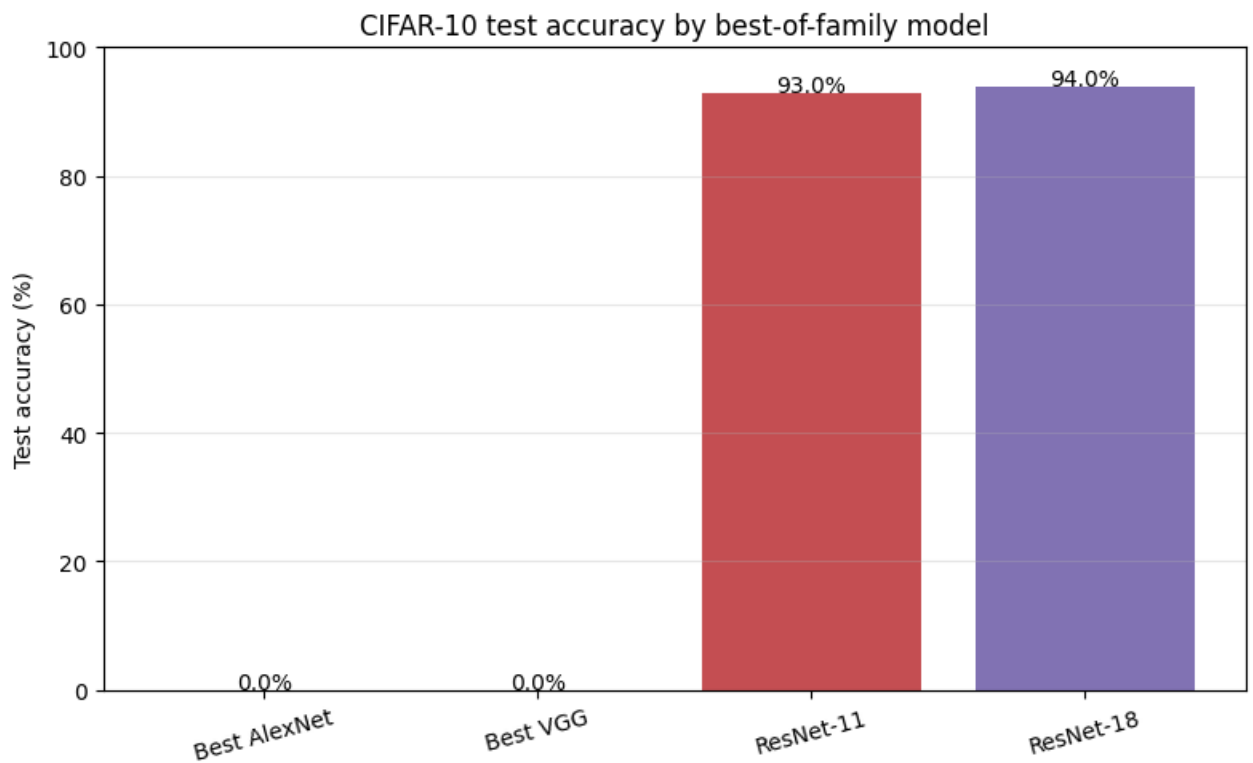
```

```

    ('ResNet-11', h_r11['params'], acc_r11),
    ('ResNet-18', h_r18['params'], acc_r18)]
print(f"{'Model':14s}{'Params':>12s}{'TestAcc':>10s}")
for n,p,a in finals: print(f"{n:14s}{p:>12,}{a*100:>9.2f}%")
names=[f[0] for f in finals]; accs=[f[2]*100 for f in finals]
plt.figure(figsize=(8,5))
bars=plt.bar(names,accs,color=['#4C72B0','#55A868','#C44E52','#8172B3'])
for b,a in zip(bars,accs): plt.text(b.get_x()+b.get_width()/2,a+0.2,f"{a*100:.2f}%")
plt.ylabel('Test accuracy (%)'); plt.title('CIFAR-10 test accuracy by best-of-family model')
plt.xticks(rotation=15); plt.ylim(0,100); plt.grid(axis='y',alpha=.3);

```

Model	Params	TestAcc
Best AlexNet	6,975,178	0.00%
Best VGG	9,231,114	0.00%
ResNet-11	9,623,882	92.97%
ResNet-18	11,173,962	93.95%



```

r34,h_r34 = train_model(resnet34(), "ResNet-34", lr=0.1)
plot_history([h_r18,h_r34], "ResNet-18 vs ResNet-34")
report_test(r34,"ResNet-34")
print(f"ResNet-34 params: {count_params(resnet34()):,}")

```

```

=== ResNet-34 | params=21,282,122 | epochs=50 | lr=0.1 ===
ep 01/50 train_loss=2.182 acc=0.214 | val_loss=1.820 acc=0.315 | 68.7
ep 02/50 train_loss=1.708 acc=0.359 | val_loss=1.581 acc=0.418 | 68.7
ep 03/50 train_loss=1.486 acc=0.447 | val_loss=1.373 acc=0.500 | 67.8
ep 04/50 train_loss=1.257 acc=0.544 | val_loss=1.128 acc=0.598 | 68.4
ep 05/50 train_loss=1.049 acc=0.625 | val_loss=1.285 acc=0.571 | 68.0
ep 06/50 train_loss=0.930 acc=0.668 | val_loss=0.985 acc=0.648 | 68.1
ep 07/50 train_loss=0.823 acc=0.708 | val_loss=0.984 acc=0.668 | 68.0
ep 08/50 train_loss=0.725 acc=0.743 | val_loss=0.717 acc=0.750 | 67.9
ep 09/50 train_loss=0.633 acc=0.780 | val_loss=0.677 acc=0.761 | 68.0
ep 10/50 train_loss=0.582 acc=0.799 | val_loss=0.753 acc=0.753 | 68.4
ep 11/50 train_loss=0.532 acc=0.817 | val_loss=0.677 acc=0.771 | 68.1
ep 12/50 train_loss=0.500 acc=0.827 | val_loss=0.630 acc=0.800 | 68.1
ep 13/50 train_loss=0.467 acc=0.839 | val_loss=0.549 acc=0.807 | 68.4
ep 14/50 train_loss=0.444 acc=0.849 | val_loss=0.519 acc=0.819 | 68.4
ep 15/50 train_loss=0.428 acc=0.853 | val_loss=0.481 acc=0.836 | 68.4
ep 16/50 train_loss=0.408 acc=0.860 | val_loss=0.511 acc=0.825 | 68.0
ep 17/50 train_loss=0.389 acc=0.867 | val_loss=0.581 acc=0.808 | 67.9
ep 18/50 train_loss=0.373 acc=0.874 | val_loss=0.489 acc=0.840 | 67.8
ep 19/50 train_loss=0.358 acc=0.876 | val_loss=0.499 acc=0.829 | 67.7
ep 20/50 train_loss=0.340 acc=0.885 | val_loss=0.724 acc=0.789 | 67.8
ep 21/50 train_loss=0.327 acc=0.889 | val_loss=0.485 acc=0.840 | 68.0
ep 22/50 train_loss=0.311 acc=0.893 | val_loss=0.430 acc=0.863 | 68.1
ep 23/50 train_loss=0.293 acc=0.898 | val_loss=0.464 acc=0.845 | 68.1
ep 24/50 train_loss=0.285 acc=0.903 | val_loss=0.467 acc=0.844 | 67.0

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